Substrate Carries, Architecture Pays: Check-In-Level Embeddings for Multi-Task POI Prediction

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Abstract. Multi-task learning for point-of-interest prediction typically pairs joint next-category and next-region heads over a shared graph embedding, assuming joint training transfers signal across tasks. We swap the per-place substrate for a check-in-level contextual graph embedding (one vector per visit) and run a five-state Gowalla ablation under user-disjoint folds with leak-free per-fold transition priors. The substrate is task-asymmetric: per-visit context lifts single-task next-category macro-F1 by 14 to 29 percentage points at every state (paired Wilcoxon p of 0.0312, the maximum at five folds), while per-place embeddings tie or marginally exceed it under matched-head ceilings on next-region. Joint multi-task training over a cross-attention backbone then delivers a category lift at four of five states; the cost is a sign-consistent 7 to 17 percentage point drop on next-region top-10 accuracy, the textbook tradeoff. A counterfactual at Alabama attributes about 72 percent of the substrate gain to per-visit context.

Keywords: Multi-Task Learning · Point-of-Interest Prediction · Graph Embeddings.

1 Introduction

Point-of-Interest (POI) prediction in Location-Based Social Networks couples two tasks over a shared substrate. Given a length-9 window of check-ins, we predict the next visit's semantic category and spatial region. The tasks place disjoint demands: **category prediction** is a 7-way decision shaped by per-visit context (intent, time-of-day, co-location); **region prediction** is a fine-grained ranking over 1.1K to 8.5K classes hinging on spatial-temporal regularities and short-horizon transitions. The MTL appeal is standard: a shared representation should let one head's signal transfer and lift both [2,19].

Two recent works frame the bottleneck. The first reported hard-parameter-sharing MTL on POI-stable graph embeddings (DGI [20]) yielding only marginal gains within standard deviations; the diagnosis was representation mismatch. A follow-up decomposed the monolithic embedding into spatial, temporal, and categorical sub-encoders, recovering category-side gains across three states. Together these suggest embedding choice, not the MTL recipe, is the load-bearing factor.

Both prior responses treat the embedding as a *what* axis. We test an orthogonal *granularity* axis: emission rate. POI-stable embeddings like HGI [7] emit one vector per place; a check-in-level variant emits one per visit, capturing intent shifts across two visits to the same café at different hours. Our hypothesis is that **per-visit context** is the load-bearing substrate property for the easier head, and that once the substrate is fixed, joint training behaves like the textbook MTL tradeoff. The carry is task-asymmetric: per-visit *carries* next-category; per-place embeddings tie or marginally exceed it on next-region.

We adopt Check2HGI (a check-in-level extension of the hierarchical graph-infomax line emitting 64-dim per-visit vectors) and run a controlled ablation on five Gowalla [5] U.S.-state splits (AL, AZ, FL, CA, TX) under user-disjoint folds with leak-free per-fold transition priors. We compare Check2HGI against HGI under matched-head STL baselines, and run a cross-attention MTL backbone (per-task input streams, GRU category head, STAN-Flow region head [12]). The contribution is empirical and methodological: we propose no new balancer, head, or attention variant, and do not claim Check2HGI as our model. We reattribute the default MTL-for-POI reading by separating substrate from architecture, and surface a cross-attention ablation pitfall.

The main contributions of this paper are as follows:

- **C1. Substrate is task-asymmetric.** A controlled five-state ablation showing that check-in-level contextual graph embeddings provide a task-asymmetric substrate for joint POI category and region prediction. Matched-head single-task category macro-F1 lifts by +14.5 to +29 percentage points (pp) at every state (paired Wilcoxon $p = 0.0312$ each, the maximum at $n = 5$ folds; 5/5 folds positive; head-invariant at AL/AZ and matched-head replicated at FL/CA/TX), while per-place embeddings tie or marginally exceed Check2HGI on next-region under matched-head ceilings (TOST non-inferiority at $\delta = 2$ pp passes at CA and TX). A pooled-vs-canonical counterfactual at Alabama attributes about 72 % of the category substrate gap to per-visit context, single-state mechanism evidence consistent with the asymmetry.
- **C2. Classic MTL tradeoff.** A direct measurement of the textbook multi-task tradeoff under the load-bearing substrate. With Check2HGI fixed, joint cross-attention MTL adds a small category lift at four of five states (AZ +1.20, FL +1.40, CA +1.68, TX +1.89 pp at $p \leq 2\text{e}{-6}$ across $n = 20$ multi-seed fold-pairs) and is small-significantly negative at Alabama ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$; magnitude under 2 % relative). On next-region it pays a sign-consistent 7 to 17 pp cost on Acc@10 at every state ($p \leq 1.9\text{e}{-6}$, $n = 20$); drop-in alternatives (FAMO, Aligned-MTL, a hierarchical-additive-softmax region head) do not close the gap, indicating that the cost is structural to the cross-attention shared-backbone interaction rather than a recipe artefact.
- A methodological note (developed in Section 6): under cross-attention MTL, a loss-side `task_weight = 0` ablation does not silence the silenced encoder, since it co-adapts through the attention key and value projections; encoder-

frozen isolation is the clean architectural decomposition, backed by regression tests in our anonymous code release.

Sections 2, 3, and 4 cover related work, method, and setup. Section 5 reports substrate ablation, MTL-vs-STL, Δ_m, and external baselines. Section 6 presents the counterfactual, drop-in ablations, and the cross-attention note. Sections 7 and 8 close.

2 Related Work

Three lines of work bear on our study: graph-based POI embeddings that supply the substrate, multi-task learning balancers that sit on top of any joint architecture, and prior multi-task work on the POI category and next-POI pair. We summarise each line briefly and position our contribution against it; a fourth note on cross-attention multi-task models is deferred to the methodological discussion in Section 6.

2.1 Graph-based POI embeddings

Substrate choice is the load-bearing axis our study tests, so we trace the graph-based line that produces it. Early POI representations such as POI2Vec [6] learn one vector per place from co-visit statistics, treating the POI as the atomic unit. CATAPE [16] extends the idea by injecting category supervision into the embedding objective, but the emission rate stays at one vector per POI. Self-supervised graph contrastive methods then sharpened the representation: Deep Graph Infomax (DGI) [20] maximises mutual information between local node embeddings and a global graph summary, and Hierarchical Graph Infomax (HGI) [7] adapts the contrast across POI, region, and city graphs to capture multi-scale urban structure. Both DGI and HGI emit *POI-stable* vectors, so the same place visited twice yields the same representation regardless of when or with whom the visit happened. The substrate we use, *Check2HGI*, is a check-in-level extension of HGI's hierarchical contrastive objective: contrastive views are sampled at the granularity of a single check-in rather than a single POI, and the visit's temporal and co-visit context perturbs the view, so the same POI visited twice can yield two distinct vectors. This per-visit emission rate is the property our substrate axis tests; the recipe is described in supplementary material and in our anonymous code release.

2.2 Multi-task learning balancers

A second body of work studies how to combine task losses or task gradients once the architecture is fixed. The line of gradient-balancing methods includes GradNorm [4], which rescales per-task gradients to match a target relative training rate, PCGrad [23], which projects each task's gradient onto the orthogonal complement of any conflicting task, CAGrad [11], which constrains the joint update to descend on every task's loss within a trust region, and NashMTL [15],

which solves a small bargaining game per step to choose the update direction. More recent proposals continue along the same axis: FAMO [10] adapts loss weights using per-task progress estimates, and Aligned-MTL [17] aligns gradient directions through an independent-component decomposition. The classic survey of multi-task learning for dense prediction [19] aggregates these methods and reports that no balancer dominates across task pairs; the textbook tradeoff identified by Caruana [2] and quantified by Maninis et al. [13] through the Δ_m joint score remains the operating regime. Our setup uses a fixed static-weight aggregation with $\lambda_{\text{cat}} = 0.75$. We do not propose a new balancer, and we ablate against FAMO and Aligned-MTL in Section 6 to check whether a drop-in balancer recovers the next-region cost we will report.

2.3 Prior POI multi-task work

A third line is concept-aligned with our task pair. HMT-GRN [9] jointly predicts next-POI and next-region through a hierarchical graph recurrent network, GETNext [22] combines a transformer with an additive log-transition prior over a trajectory flow map, and the human-mobility representation model [3] jointly models attribute and trajectory signals through shared encoders. STAN [12] addresses the next-location subproblem with spatio-temporal self-attention and is the backbone we adapt for our region head; we borrow GETNext's additive $\alpha \cdot \log T$ prior pattern but do not include its friendship or check-in graph priors, so our region head is not a faithful GETNext reproduction. ReHDM [8] extends the hypergraph line with a region-aware dual-level model and serves as our concept-aligned external next-region baseline. Two recent works on the precise category and next-POI pairing are closer still. A first study trained a hard-parameter-sharing multi-task framework over POI-stable graph embeddings (DGI), comparing against single-task baselines such as HMRM [3] and MHA+PE [24], and reported only marginal joint gains, with most differences within standard deviations; the diagnosis pointed at representation mismatch, where a single shared encoder learns a compromise specialised enough for neither head. A follow-up tested an embedding-side response by decomposing the monolithic graph embedding into independent spatial, temporal, and categorical sub-encoders, and recovered substantial category-side gains across three states, supporting the reading that the embedding choice was the load-bearing factor. Both responses kept the embedding's per-POI emission rate fixed and varied the *what* axis (which features to encode); our study varies the *granularity* axis instead.

Our work re-examines a default reading of multi-task learning for POI category and next-POI prediction: that joint training transfers signal between heads to lift both jointly. With the substrate held fixed at the right granularity, we find that joint cross-attention training delivers a small additional category lift at four of five Gowalla [5] state splits and pays a sign-consistent cost on next-region at every state, the textbook tradeoff. Our contribution is empirical and methodological. We do not propose a new balancer, head, or attention variant;

we use a controlled five-state ablation on user-disjoint folds with leak-free per-fold transition priors to reattribute the joint result, separating the substrate's contribution from the architecture's.

3 Method

We describe the substrate (adopted), the backbone (the object of study), and the training protocol. Notation: a check-in is $c = (p, r, t)$ with POI p, region r, timestamp t; trajectories split into non-overlapping length-nine windows predicting the category and region of the tenth check-in ($L = 9$).

3.1 The Check2HGI substrate

The substrate extends the hierarchical graph-infomax line at the check-in level. HGI [7], building on Deep Graph Infomax [20], contrasts local node embeddings against a global graph summary across POI, region, and city graphs and emits one vector per place. *Check2HGI* ports the same hierarchical contrastive objective to the check-in graph: positive views are sampled per check-in, so temporal context and co-visit neighbours perturb the contrastive view. The same POI visited twice can yield two distinct 64-dimensional vectors; per-visit emission rate distinguishes the substrate from POI-stable alternatives like DGI and HGI.

Trained once per state and frozen, the substrate emits 64-dimensional vectors as a length-nine sequence, drop-in compatible with HGI. Two products feed the backbone: a per-check-in vector $\mathbf{e}_c \in \mathbb{R}^{64}$ for the category stream, and a per-region vector $\mathbf{e}_r \in \mathbb{R}^{64}$ from the region-graph level for the region stream. Both share one training run. We treat *Check2HGI* as a fixed input substrate, not a method contribution.

3.2 Cross-attention multi-task backbone

The backbone keeps two parallel input streams (one per task) and bridges them with cross-attention before a shared backbone splits into per-task heads. Let $X^c, X^r \in \mathbb{R}^{L \times d}$ feed the category and region sides at $L = 9$, $d = 64$. Two task-specific encoders (two-layer MLPs with LayerNorm and dropout) project each stream to $\mathbb{R}^{L \times d_m}$ at $d_m = 256$, yielding H^c and H^r. Single-modality variants collapsed one head in development, so two streams are the default.

A cross-attention block bridges the streams. With $\mathrm{MHA}(Q, K, V)$ scaled dot-product attention ($h = 8$),

$$\widetilde{H}^c = H^c + \mathrm{MHA}(H^c,\, H^r,\, H^r), \qquad \widetilde{H}^r = H^r + \mathrm{MHA}(H^r,\, H^c,\, H^c), \tag{1}$$

so each stream queries the other. The cross-attended streams then pass through a shared backbone of four residual blocks (LayerNorm, linear, LeakyReLU, dropout) before splitting into per-task heads (Figure 1). Eq. (1) is the architectural object the result tables interrogate.

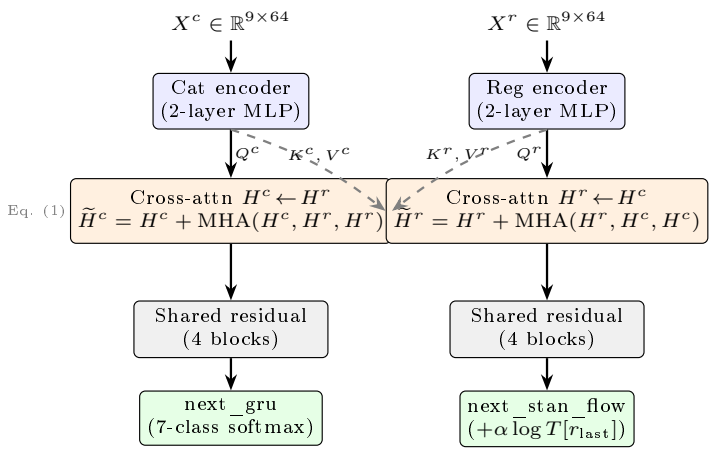

Fig. 1. Cross-attention multi-task backbone. Two task-specific encoders project per-task input streams X^c, X^r into H^c, H^r. A symmetric cross-attention block (Eq. 1) lets each stream query the other through key/value projections of the surviving stream; a four-block shared residual backbone then splits into the `next_gru` category head and the `next_stan_flow` region head. Dashed arrows mark the K/V cross-feeds responsible for the encoder co-adaptation discussed in Section 6.3.

The category head, `next_gru`, is a two-layer GRU over the cross-attended check-in stream with attention pooling and seven-class softmax. The region head, `next_stan_flow`, is an additive-prior STAN [12] variant: a STAN backbone over the cross-attended region stream, scored against learnable region embeddings, with an additive log-transition prior $\alpha \cdot \log T[r_{\mathrm{last}}]$ (learnable α, built per fold from training-fold edges only, so leak-free). The log-prior pattern follows GETNext, but our head omits friendship and check-in graph priors and is not a faithful reproduction. STL ceilings reuse these head classes, so MTL and STL rows differ only in the backbone.

3.3 Training protocol

We aggregate the two task losses with a static weighting,

$$\mathcal{L} = \lambda_{\mathrm{cat}}\, \mathcal{L}_{\mathrm{cat}} + \lambda_{\mathrm{reg}}\, \mathcal{L}_{\mathrm{reg}}, \qquad \lambda_{\mathrm{cat}} = 0.75, \quad \lambda_{\mathrm{reg}} = 0.25, \tag{2}$$

where $\mathcal{L}_{\mathrm{cat}}, \mathcal{L}_{\mathrm{reg}}$ are unweighted cross-entropy, fixed across all MTL rows. FAMO [10] and Aligned-MTL [17] drop-in comparisons are in Section 6.

The optimiser is AdamW with three parameter groups (category, region, shared), OneCycleLR, batch size 2048 (1024 at the largest state), gradient accumulation of two. Training runs 50 epochs in FP32 with no autocast; per-task best checkpoints select on validation Acc@10 (region) and macro-F1 (category) for epochs at least five, since earlier checkpoints saturate the region head before the category head converges. Class weights are applied to neither head: weighted

Table 1. Dataset statistics for the five Gowalla [5] U.S.-state splits, after filtering to users with at least five check-ins. The headline three (FL/CA/TX) sit above the rule; the smaller-scale anchors (AL/AZ) sit below. Region cardinality is the prediction target on the next-region task.

State	Users	Check-ins	POIs	Regions	Mean traj. length
Florida (FL)	13,935	1,392,262	76,266	4,703	99.9
California (CA)	26,171	3,148,594	168,771	8,501	120.3
Texas (TX)	27,603	4,067,543	160,575	6,553	147.4
Alabama (AL)	1,622	109,695	11,666	1,109	67.6
Arizona (AZ)	3,331	227,956	20,469	1,547	68.4

CE on category pushed gradient mass against region, so both use unweighted CE, matching the STL ceilings of Section 4. Two parameter-group recipes are scale-conditional in our search; selection is in Section 6.

Folds, seeds, and multi-seed pooling follow the user-disjoint StratifiedGroupKFold setup of Section 4; the same splits are reused across substrates, heads, and recipes so paired tests are well defined. A methodological caveat: under cross-attention, a loss-side ablation that zeros one task's weight does not isolate the silenced encoder, since it co-adapts through the attention key and value projections of the surviving stream. We use encoder-frozen isolation (random-init, frozen) for clean architectural decomposition; the protocol is in Section 6. Code, fold splits, seeds, per-fold transition priors, and regression tests are released at `https://anonymous.4open.science/r/PoiMtlNet-FE6A/`.

4 Experimental Setup

4.1 Datasets and preprocessing

We evaluate on five U.S.-state splits of the public *Gowalla* check-in dataset [5]: AL, AZ, FL, CA, TX. Each split is filtered to users with at least five check-ins, applied per state before fold creation so each state preserves its native cardinality. Per-state statistics are in Table 1; FL/CA/TX serve as the headline scale and AL/AZ as smaller-scale anchors. Region labels come from the per-state region partition used by the substrate engine.

For each user we form non-overlapping length-9 windows over the chronological sequence and predict the category and region of the tenth check-in (shorter sequences padded). Folds are **user-disjoint** via `StratifiedGroupKFold(groups=userid, shuffle=True, random_state=42)` with five folds, so train/val/test never share a user. Splits are reused across substrates, heads, and recipes so paired tests are well defined. The next-region transition prior is rebuilt per fold from training-fold edges only and seed-tagged, so the leak-free protocol holds at every comparison.

4.2 Baselines and metrics

The headline contrast is the cross-attention MTL model vs. matched-head STL ceilings: `next_gru` for category and `next_stan_flow` (STAN [12] backbone with additive log-transition prior over training-fold-only edges) for region. Matching head classes isolates the architectural contribution of joint training from head capacity. External baselines are POI-RGNN, MHA+PE (category) and Markov-1-region, STL STAN, ReHDM (region); these are reported in §5.

Category metrics: macro-F1 (primary), Acc@1/Acc@3 (secondary), since the 7-way distribution is skewed. Region metrics: Acc@10 (primary), Acc@5, MRR (secondary). The joint score is Δ_m [13,19] against the matched-head STL ceiling, reported as primary (macro-F1, MRR) and secondary (macro-F1, Acc@10).

4.3 Statistical protocol

Paired contrasts use the **paired Wilcoxon signed-rank** test [21] with `alternative='greater'` for directional claims and the two-one-sided test (**TOST**) for non-inferiority at $\delta \in \{2, 3\}$ pp. At the single-seed five-fold scale the max significance for $n = 5$ is $p = 0.0312$ (all folds agreeing).

For the headline MTL-vs-STL contrast we pool across four extra seeds $\{0, 1, 7, 100\}$ on top of seed-42, yielding $n = 20$ multi-seed fold-pairs per state. The FL Δ_m axis additionally carries $n = 25$ fold-pairs from a second sweep, used to mark the strongest paired result. Effect sizes are in pp for F1/Acc@K/MRR; Δ_m is in percent. Code, fold splits, seeds, and per-fold priors are in the anonymous snapshot referenced in §3.

5 Results

We report the empirical reattribution across five Gowalla state splits. Section 5.1 isolates the substrate axis (Check2HGI vs HGI under matched-head STL ceilings). Section 5.2 compares the cross-attention MTL backbone against those ceilings, the headline contrast. Section 5.3 reports the joint Δ_m score. Section 5.4 situates the model against external baselines.

5.1 Substrate ablation: per-visit context is task-asymmetric

Table 2(a) reports next-category macro-F1 under the matched-head `next_gru` STL ceiling, with substrate as the only changing factor. Check2HGI exceeds HGI by +14.52 to +29.02 percentage points (pp) at every state (smallest at AZ, largest at FL). The paired Wilcoxon signed-rank test reaches the $n = 5$ ceiling everywhere ($p = 0.0312$, 5/5 folds positive). Two robustness probes back this: the AL/AZ gap is head-invariant across linear, single-layer, GRU, and LSTM probes, and the FL/CA/TX gap replicates under the matched-head ceiling. We read the lift as evidence that per-visit context, not POI-level attribute encoding, is what the next-category head exploits.

Table 2. Substrate ablation under matched-head single-task ceilings, five-fold cross-validation. Top: next-category macro-F1, head `next_gru`, $n = 5$ Wilcoxon ceiling (HGI single-seed; Check2HGI at AL/AZ/FL is multi-seed mean $\pm$ seed σ over seeds $\{0, 1, 7, 100\}$); the asterisk marks 5/5 folds positive at $p = 0.0312$ one-sided. Bottom: next-region Acc@10, head `next_stan_flow` with the additive log-transition prior built per fold from training-fold edges only; negative Δ values indicate POI-stable HGI is nominally ahead. The TOST columns report two-one-sided non-inferiority verdicts at $\delta = 2$ and $\delta = 3$ pp: ✓ indicates the gap stays inside the equivalence margin, × that it does not.

(a) Next-category macro-F1.

State	HGI cat F1	Check2HGI cat F1	Δ (pp)	Wilcoxon p
Alabama (AL)	25.26 ± 1.18	41.35 ± 0.17	+15.50*	0.0312
Arizona (AZ)	28.69 ± 0.79	43.90 ± 0.17	+14.52*	0.0312
Florida (FL)	34.41 ± 1.05	63.43 ± 0.98	+29.02*	0.0312
California (CA)	31.13 ± 1.04	59.94 ± 0.59	+28.81*	0.0312
Texas (TX)	31.89 ± 0.55	60.24 ± 1.84	+28.34*	0.0312

(b) Next-region Acc@10.

State	HGI Acc@10	Check2HGI Acc@10	Δ (pp)	TOST $\delta = 2$	TOST $\delta = 3$
Alabama (AL)	61.86 ± 3.29	59.15 ± 3.48	−2.71	×	×
Arizona (AZ)	53.37 ± 2.55	50.24 ± 2.51	−3.13	×	×
Florida (FL)	71.34 ± 0.64	69.22 ± 0.52	−2.12	×	✓
California (CA)	57.77 ± 1.12	55.92 ± 1.20	−1.85	✓	✓
Texas (TX)	60.47 ± 1.26	58.89 ± 1.28	−1.59	✓	✓

Table 2(b) reports Acc@10 under the matched-head `next_stan_flow` STL ceiling. POI-stable HGI is nominally above Check2HGI by 1.6 to 3.1 pp at every state; on FL/CA/TX the gap shrinks to 1.6 to 2.1 pp and TOST non-inferiority at $\delta = 2$ pp passes at CA and TX. At FL the $\delta = 3$ pp test passes. AL and AZ stay outside both margins. The short-horizon transition structure driving the region head is captured equally well, or fractionally better, by per-place embeddings whose category-level pooling smooths the per-visit variance that helps the category head.

The substrate's effect is task-asymmetric. The category head reads per-visit context and lifts by an order of magnitude more than its standard-deviation envelope; the region head reads transition structure preserved under per-place pooling, tying within ±2 pp. Section 6 attributes the asymmetry through a per-visit-context counterfactual at Alabama. Any MTL system on this task pair operates on a substrate that already encodes the easier task more clearly than the harder one.

5.2 Multi-task vs single-task on the load-bearing substrate

Table 3 reports the headline contrast: the cross-attention MTL backbone against the matched-head `next_gru` and `next_stan_flow` STL ceilings, both on Check2HGI,

Table 3. Headline MTL-vs-STL contrast on the Check2HGI substrate, multi-seed pooled at $n = 20$ fold-pairs per state (seeds $\{0, 1, 7, 100\}$, five user-disjoint folds). The MTL row is the cross-attention multi-task backbone; the STL rows are matched-head single-task ceilings, `next_gru` for category and `next_stan_flow` for region. Cells are mean $\pm$ pooled σ over $n = 20$ fold-pairs. The Δ columns report MTL $-$ STL in percentage points; the Wilcoxon p-values are paired one-sided at $n = 20$ (alternative "greater" for category gains, "less" for region drops where the headline direction is negative). FL/CA/TX sit above the rule as the headline scale; AL/AZ sit below as small-scale anchors.

	Next-category (macro-F1)			Next-region (Acc@10)		
State	STL	MTL	Δ (pp), p	STL	MTL	Δ (pp), p
Florida (FL)	67.16 ± 0.13	68.56 ± 0.79	+1.40, 2e−6	70.62 ± 0.09	63.27 ± 0.10	−7.34, 1.9e−6
California (CA)	62.39 ± 0.13	64.07 ± 0.14	+1.68, 2e−6	56.85 ± 0.09	47.35 ± 0.11	−9.50, 1.9e−6
Texas (TX)	63.11 ± 0.13	65.00 ± 0.11	+1.89, 2e−6	59.44 ± 0.09	42.84 ± 0.14	−16.59, 1.9e−6
Alabama (AL)	41.35 ± 0.17	40.57 ± 0.24	−0.78, 0.036	61.21 ± 0.18	50.17 ± 0.24	−11.04, 1.9e−6
Arizona (AZ)	43.90 ± 0.17	45.10 ± 0.19	+1.20, <1e−4	53.06 ± 0.15	40.78 ± 0.07	−12.27, 1.9e−6

multi-seed pooled at $n = 20$ fold-pairs per state. Category is positive at four of five states: FL +1.40 pp (p = 2e−6), CA +1.68 pp (p = 2e−6), TX +1.89 pp (p = 2e−6), AZ +1.20 pp (p < 1e−4). At AL the contrast is small-significantly negative ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$), under 2 % relative on the 41 % F1 scale. The four positive states sit outside the seed and fold envelope; we report the AL negative as is rather than absorbing it.

Next-region is the textbook tradeoff. MTL pays −7.34 pp (FL) to −16.59 pp (TX) on Acc@10 at every state, all paper-grade significant ($p \leq$ 1.9e−6). The pattern is broadly downward on AL→AZ→FL, with CA preserving the FL regime and TX breaking the monotonic shrink. We read the TX non-monotonicity descriptively: state-specific factors beyond raw class count (geographic POI distribution, user-density skew, per-state log-transition prior coverage) plausibly contribute; we do not read the AL→FL contraction as an inferential scaling claim.

With the substrate fixed at the per-visit granularity, the MTL backbone delivers a small category lift at four of five states and pays a sign-consistent cost on the harder head at every state, the classic MTL tradeoff. The cost is structural to the cross-attention shared-backbone interaction with the region head; Section 6 reports drop-in replacements (FAMO, Aligned-MTL, hierarchical-additive-softmax region head) that fail to close the next-region gap.

5.3 Joint Δ_m score: a Pareto-positive cell at Florida

We compute Δ_m [13,19] against the matched-head STL ceiling. On the primary pair (category macro-F1, region MRR), FL carries an $n = 25$ multi-seed sweep and is the strongest paired result in the paper: Δ_m-MRR = +2.33 % at p = 2.98e−8, 25/25 fold-pairs positive. The secondary pair at FL is Δ_m-Acc@10 = −1.12 % at p = 3.20e−5 (4/25); MRR rewards near-correct candidates that raw

Table 4. External-baseline comparison at the headline scale (FL, CA, TX). The category block reports macro-F1; the region block reports Acc@10. Italicised STL rows are matched-head single-task ceilings on the Check2HGI substrate (`next_gru` for category; `next_stan_flow` with the per-fold additive log-transition prior for region); the bold MTL row is our cross-attention multi-task model. POI-RGNN and MHA+PE numbers are our user-disjoint reproductions; the original POI-RGNN evaluation [1] used non-user-disjoint folds, so the absolute values are not directly comparable. ReHDM and faithful STAN faithful runs are reported at AL/AZ/FL only; CA/TX cells are deferred to the camera-ready (the dual-level hypergraph in ReHDM scales quadratically with region cardinality, exceeding our compute budget at 8.5K and 6.5K regions respectively). Per-state cells are mean $\pm\sigma$ over the five user-disjoint folds.

	Next-category macro-F1			Next-region Acc@10		
Method	FL	CA	TX	FL	CA	TX
External baselines (faithful reproductions)						
Markov-1 (floor)	23.98 ± 1.22	19.97 ± 1.77	18.17 ± 1.38	65.05 ± 0.93	52.09 ± 0.80	54.94 ± 0.46
POI-RGNN [1]	34.49 ± 1.14	31.78 ± 0.82	33.03 ± 0.70	–	–	–
MHA+PE [24]	32.06 ± 0.23	29.13 ± 0.71	29.91 ± 0.43	–	–	–
STAN [12] (HGI)	–	–	–	73.58 ± 0.43	60.45 ± 0.97	62.70 ± 0.37
ReHDM [8]	–	–	–	65.68 ± 0.26	–†	–†
Our matched-head single-task ceilings (Check2HGI substrate)						
STL `next_gru` (cat ceiling)	67.16 ± 0.13	62.39 ± 0.13	63.11 ± 0.13	–	–	–
STL `next_stan_flow` (reg ceiling)	–	–	–	70.62 ± 0.09	56.85 ± 0.09	59.44 ± 0.09
Proposed cross-attention multi-task model						
MTL (ours)	**68.56 ± 0.79**	**64.07 ± 0.14**	**65.00 ± 0.11**	63.27 ± 0.10	47.35 ± 0.11	42.84 ± 0.14

† ReHDM at CA/TX deferred (camera-ready, compute budget).

top-10 accuracy misses. The other four states report at the $n = 5$ Wilcoxon ceiling: Δ_m is negative on both axes (-24.84 to -1.61 % MRR; -22.41 to -6.85 % Acc@10), at most 1/5 fold-pairs positive. FL is the lone Pareto-positive cell on the primary axis; the other four are reported under the $n = 5$ ceiling caveat.

5.4 External baselines

Table 4 situates the model against external baselines. On next-category, our matched-head `next_gru` STL ceiling on Check2HGI exceeds the faithful POI-RGNN reproduction [1] by +27.4 pp (FL: 67.16 vs 34.49) to +30.6 pp (CA: 62.39 vs 31.78); our MTL row widens the gap to +30.0 to +33.2 pp. Faithful MHA+PE [24] sits below POI-RGNN at all three states (FL 32.06, CA 29.13, TX 29.91). Two qualifications: published POI-RGNN [1] used non-user-disjoint folds, so absolute gaps to the original numbers are not comparable, and our reproduction enforces user-disjoint folds. We do not read the external category gap as a state-of-the-art claim; the substrate-axis evidence attributes the bulk of it to per-visit context, not MTL aggregation.

On next-region, the faithful STAN reproduction [12] on HGI is the strongest external baseline at all three states (FL 73.58, CA 60.45, TX 62.70 Acc@10), exceeding our STL `next_stan_flow` ceiling on Check2HGI (FL 70.62, CA 56.85,

TX 59.44). With the substrate-ablation reading, POI-stable HGI is a more natural fit for the region head, and the per-visit substrate's region cost reflects the same property that helps the category head. Faithful ReHDM [8] at FL (65.68 Acc@10) sits below both STAN-on-HGI and our `next_stan_flow` ceiling, and below the per-fold Markov-1-region floor at AZ. Our MTL row trails STAN-on-HGI on Acc@10 by −10.31 pp (FL: 63.27 vs 73.58) to −19.86 pp (TX: 42.84 vs 62.70), consistent with the substrate gap compounded by the cross-attention reg cost.

The category gap to external baselines is large and substrate-driven; the region picture favours POI-stable embeddings under a matched-head STAN backbone, and the MTL backbone on the per-visit substrate pays a structural region cost that drop-in MTL recipes do not close (Section 6).

6 Mechanism and Ablations

This section turns the reattribution into a mechanism reading, stress-tests the next-region cost against drop-in alternatives, and closes with a methodological note. Section 6.1 isolates per-visit context at one state. Section 6.2 reports adaptive balancers and a head-side fix that fail to close the gap. Section 6.3 states the cross-attention loss-side pitfall and the encoder-frozen protocol that decomposes it cleanly.

6.1 Per-visit context counterfactual at Alabama

We isolate per-visit context through a pooled-vs-canonical counterfactual at Alabama. POI-pooled *Check2HGI* mean-pools canonical vectors per POI across all check-ins and applies the per-place result uniformly to every visit. This kills per-visit variation while preserving the training signal: canonical vs. pooled isolates per-visit context, pooled vs. HGI isolates the contrastive training signal.

Under the matched-head `next_gru` STL ceiling at AL, canonical Check2HGI reaches 40.76 macro-F1, POI-pooled reaches 29.57, HGI reaches 25.26 (Figure 2). The per-visit component is +11.19 pp (∼72 % of the substrate gap), the training-signal component +4.31 pp (∼28 %). The split survives a head-free linear probe (30.84 vs. 23.20 vs. 18.70, per-visit share ∼63 %); the GRU head amplifies the per-visit share rather than creating it. Per-visit variance is what the category head exploits; pooling to per-place vectors (HGI) or to region cardinality (region head) smooths the same variance away, consistent with the asymmetry in Section 5.1.

This is single-state evidence; we do not generalise the ∼72 % / ∼28 % split to the other four states, and a second-state replication is left to future work. The mechanism nonetheless explains *why* the substrate carries the easier head and trails on the harder one: what it adds at AL is exactly what the region head does not need.

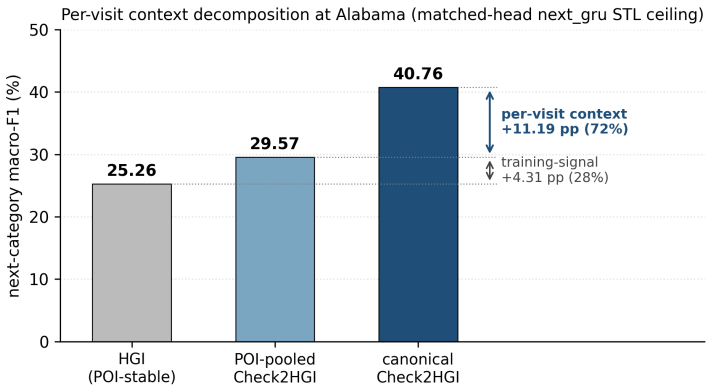

Fig. 2. Per-visit context decomposition at Alabama under the matched-head `next_gru` single-task ceiling. The substrate gap (canonical Check2HGI − HGI = +15.50 pp) splits into a per-visit-context component (canonical − POI-pooled = +11.19 pp, ∼72 %) and a training-signal component (POI-pooled − HGI = +4.31 pp, ∼28 %). The bulk of the gap comes from per-visit variance, the property POI-stable embeddings cannot supply and that pooling to region cardinality smooths away.

6.2 Drop-in MTL ablations do not recover the next-region cost

The next-region cost is sign-consistent across all five states (7 to 17 pp on Acc@10). We tested whether a drop-in replacement on the MTL backbone closes the gap. We picked FL (smallest cost, easiest to recover) and ran three replacements against the same folds and seeds: a magnitude balancer (FAMO [10]), a direction-aligning balancer (Aligned-MTL [17]), and a head-side fix (hierarchical-additive-softmax region head). Each is drop-in: substrate, backbone, optimiser, and schedule fixed at Section 3.

All three miss a +3 pp recovery target on Acc@10 at FL. FAMO closes +0.62 pp, within one fold-pair sd of zero and not paired-Wilcoxon significant. Aligned-MTL ties at −0.11 pp on next-region and regresses −0.90 pp on next-category. The hierarchical-additive-softmax head reaches −3.01 pp mean across five folds; the mean is dragged by a single fold-init that collapsed to an epoch-three optimum, but even dropping that fold the recovery sits at $\approx +1.7$ pp, still below target. None reaches paired-Wilcoxon significance at $p_{\text{greater}} < 0.10$. The cost is therefore structural to the cross-attention shared-backbone interaction with the region head, not a recipe artefact: a magnitude balancer, a direction aligner, and a head-capacity fix all leave it essentially intact. The panel was run at FL only; the cost is sign-consistent across all five states, so the structural reading transfers to the panel that pays it.

6.3 Cross-attention $\lambda = 0$ is not encoder isolation

A natural ablation zeros one task's weight in Eq. (2), on the reading that $\lambda_{\text{cat}} = 0$ silences the category encoder and $\lambda_{\text{reg}} = 0$ silences the region encoder. Under

cross-attention this is unsound: the silenced encoder updates through the cross-attention key/value projections of the surviving stream. With $\lambda_{\text{cat}} = 0$ the region loss depends on the category-encoder output through the attention block, so gradient flow continues into the category encoder and its parameters co-adapt to the region task; symmetrically in the other direction. We therefore use **encoder-frozen isolation** (silenced encoder random-init, frozen) as the clean architectural decomposition; regression tests are in the anonymous snapshot. The note generalises to any cross-task attention MTL model that ablates by zeroing a task weight.

7 Discussion and Limitations

We synthesise Sections 5 and 6 into four readings: substrate task-asymmetry, scale observations, limitations, and follow-up directions.

The headline reading is that the MTL picture is dominated by the *substrate's task-asymmetric value*, not cross-task transfer through joint training. Per-visit context is paper-grade significant for next-category at every state (+14.5 to +29 pp matched-head); on next-region the same substrate ties or marginally trails HGI by 1.6 to 3.1 pp under the matched-head `next_stan_flow` ceiling, with TOST $\delta = 2$ pp passing at CA and TX. With Check2HGI fixed, the cross-attention MTL backbone adds a small category lift at four of five states (AZ +1.20, FL +1.40, CA +1.68, TX +1.89 pp at $p \leq 2\text{e}{-6}$, $n = 20$) and is small-significantly negative at AL ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$); on next-region it pays a sign-consistent 7 to 17 pp cost on Acc@10 at every state ($p \leq 1.9\text{e}{-6}$, $n = 20$). The conventional MTL framing where joint training transfers signal to lift both heads is empirically vacuous on the harder task: the category lift survives the substrate ablation, and the region cost survives the substrate fix. The honest reattribution is substrate first (task-asymmetric), MTL coupling second (tradeoff-paying).

The reg cost varies in magnitude but not sign: AL, AZ, FL trend downward (-11.04, -12.27, -7.34 pp), CA preserves at -9.50 pp, TX breaks monotonicity at -16.59 pp. We report descriptively, not as an inferential scaling claim. Trajectory geometry, transition-graph density, and short-horizon Markov coverage are plausible drivers, none directly testable at our state count.

Five limitations apply. The per-visit mechanism is single-state (AL only); the $\sim 72\,\%/\sim 28\,\%$ split is scoped to that state, and a second-state replication is left to follow-up. The drop-in MTL ablation panel ran at FL only; the cost is sign-consistent across the five states, so the structural reading transfers, but a five-state replication is left to follow-up. External baseline coverage is asymmetric: ReHDM [8] is reported at AL, AZ, FL only, because its dual-level hypergraph collaborator pool grows quadratically with region cardinality and exceeded our compute budget at CA/TX on a single H100. Our POI-RGNN [1] reproduction enforces user-disjoint folds (the published evaluation used non-user-disjoint folds); the +27 to +29 pp matched-head Check2HGI minus POI-RGNN gap is a conservative lower bound. Finally, `StratifiedGroupKFold` with shuf-

fling produces different splits across sklearn versions; paired tests within one environment are unaffected, but absolute leak-magnitude attribution requires single-environment re-runs.

Two directions follow. Asymmetric routing (PLE [18], Cross-Stitch [14]) frees the region head from shared-backbone interference; whether such routing recovers the MTL-over-STL gap on next-region without sacrificing the category lift is the next experiment. Encoder enrichment is the second axis: Paiva et al. (2026), which decomposes the monolithic embedding into spatial, temporal, and categorical sub-encoders, and our substrate response are complementary, and a fused substrate combining per-visit context with explicit spatial and temporal encoders is plausibly the path to a Pareto-positive joint solution. Both are out of scope here.

8 Conclusion

We tested whether check-in-level contextual graph embeddings carry the multi-task POI prediction win on next-category and next-region across five U.S.-state Gowalla splits. The substrate is task-asymmetric: per-visit context lifts matched-head single-task next-category macro-F1 by +14.5 to +29 pp at every state (paired Wilcoxon $p = 0.0312$, the $n = 5$ maximum); on next-region the same substrate ties or marginally trails per-place HGI by 1.6 to 3.1 pp under the matched-head `next_stan_flow` ceiling, with TOST non-inferiority at $\delta = 2$ pp passing at CA and TX. A pooled-vs-canonical counterfactual at Alabama attributes about 72 % of the category substrate gap to per-visit context, single-state mechanism evidence consistent with the asymmetry. With Check2HGI fixed, the cross-attention multi-task backbone adds a small category lift at four of five states (AZ +1.20, FL +1.40, CA +1.68, TX +1.89 pp at $p \leq 2\text{e}{-6}$ across $n = 20$ fold-pairs) and is small-significantly negative at AL ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$); on next-region it pays a sign-consistent 7 to 17 pp cost on Acc@10 at every state ($p \leq 1.9\text{e}{-6}$, $n = 20$), the textbook multi-task tradeoff. On the joint Δ_m axis, FL is the lone Pareto-positive cell on the primary (category macro-F1, region MRR) pair at +2.33 % ($p = 2.98\text{e}{-8}$, 25/25 fold-pairs positive); the other four states sit below zero on both axes at the $n = 5$ ceiling. Drop-in alternatives (FAMO, Aligned-MTL, a hierarchical-additive-softmax region head) do not close the next-region gap, and the methodological note of Section 6.3 generalises beyond our backbone: encoder-frozen isolation is the clean architectural decomposition under cross-attention.

Three forward-looking lines suggest themselves. Asymmetric routing through PLE [18] or Cross-Stitch Networks [14] could free the region head from shared-backbone interference. Encoder enrichment, fusing per-visit context with explicit spatial and temporal sub-encoders, plausibly reaches a substrate that the joint head can read without sacrificing either task. A region-side dynamic prior, conditioned on the user's recent trajectory, would target the next-region cost at the head rather than at the substrate. Recovering the next-region performance

without sacrificing the next-category lift is the open challenge this paper leaves to follow-up work.

References

1. Capanema, C.G.S., de Oliveira, G.S., Silva, F.A., Silva, T.R.M.B., Loureiro, A.A.F.: Combining recurrent and graph neural networks to predict the next place's category. Pervasive and Mobile Computing **86**, 101687 (2022). https://doi.org/10.1016/j.pmcj.2022.101687
2. Caruana, R.: Multitask learning. Machine Learning **28**(1), 41–75 (1997). https://doi.org/10.1023/A:1007379606734
3. Chen, M., Zhao, Y., Yu, X., Zheng, K.: Modeling spatial trajectories with attribute representation learning. IEEE Transactions on Knowledge and Data Engineering **34**(4), 1902–1914 (2022). https://doi.org/10.1109/TKDE.2020.3001025
4. Chen, Z., Badrinarayanan, V., Lee, C., Rabinovich, A.: GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In: Proceedings of the 35th International Conference on Machine Learning (ICML). pp. 794–803 (2018)
5. Cho, E., Myers, S.A., Leskovec, J.: Friendship and mobility: User movement in location-based social networks. In: Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD). pp. 1082–1090. ACM (2011). https://doi.org/10.1145/2020408.2020579
6. Feng, S., Cong, G., An, B., Chee, Y.M.: POI2Vec: Geographical latent representation for predicting future visitors. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 31 (2017)
7. Huang, W., Zhang, D., Mai, G., Guo, X., Cui, L.: Learning urban region representations with POIs and hierarchical graph infomax. ISPRS Journal of Photogrammetry and Remote Sensing **196**, 134–145 (2023). https://doi.org/10.1016/j.isprsjprs.2022.11.021
8. Li, Gu, Yao, Zhou, Zhu, Zhao, Du: Beyond individual and point: Next POI recommendation via region-aware dynamic hypergraph with dual-level modeling. In: Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI). p. 0343 (2025)
9. Lim, N., Hooi, B., Ng, S., Goh, Y.L., Weng, R., Tan, R.: Hierarchical multi-task graph recurrent network for next POI recommendation. In: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 1133–1143 (2022). https://doi.org/10.1145/3477495.3531989
10. Liu, B., Feng, Y., Stone, P., Liu, Q.: FAMO: Fast adaptive multitask optimization. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 36 (2023)
11. Liu, B., Liu, X., Jin, X., Stone, P., Liu, Q.: Conflict-averse gradient descent for multi-task learning. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 34 (2021)
12. Luo, Y., Liu, Q., Liu, Z.: STAN: Spatio-temporal attention network for next location recommendation. In: Proceedings of the Web Conference 2021 (WWW). pp. 2177–2185 (2021). https://doi.org/10.1145/3442381.3449998
13. Maninis, K.K., Radosavovic, I., Kokkinos, I.: Attentive single-tasking of multiple tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1851–1860 (2019)

14. Misra, I., Shrivastava, A., Gupta, A., Hebert, M.: Cross-stitch networks for multi-task learning. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 3994–4003 (2016). https://doi.org/10.1109/CVPR.2016.434
15. Navon, A., Shamsian, A., Achituve, I., Maron, H., Kawaguchi, K., Chechik, G., Fetaya, E.: Multi-task learning as a bargaining game. In: Proceedings of the 39th International Conference on Machine Learning (ICML) (2022), arXiv:2202.01017
16. Rahmani, H.A., Aliannejadi, M., Zadeh, R.M., Baratchi, M., Afsharchi, M., Crestani, F.: Category-aware location embedding for point-of-interest recommendation. In: Proceedings of the 2019 ACM SIGIR International Conference on Theory of Information Retrieval (ICTIR). pp. 173–176 (2019). https://doi.org/10.1145/3341981.3344240
17. Senushkin, D., Patakin, N., Kuznetsov, A., Konushin, A.: Independent component alignment for multi-task learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 20083–20093 (2023)
18. Tang, H., Liu, J., Zhao, M., Gong, X.: Progressive layered extraction (PLE): A novel multi-task learning (MTL) model for personalized recommendations. In: Proceedings of the 14th ACM Conference on Recommender Systems (RecSys). pp. 269–278 (2020). https://doi.org/10.1145/3383313.3412236
19. Vandenhende, S., Georgoulis, S., Gansbeke, W.V., Proesmans, M., Dai, D., Gool, L.V.: Multi-task learning for dense prediction tasks: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **44**(7), 3614–3633 (2022). https://doi.org/10.1109/TPAMI.2021.3054719
20. Veličković, P., Fedus, W., Hamilton, W.L., Liò, P., Bengio, Y., Hjelm, R.D.: Deep graph infomax. In: Proceedings of the 7th International Conference on Learning Representations (ICLR) (2019)
21. Wilcoxon, F.: Individual comparisons by ranking methods. Biometrics Bulletin **1**(6), 80–83 (1945). https://doi.org/10.2307/3001968
22. Yang, S., Liu, J., Zhao, K.: GETNext: Trajectory flow map enhanced transformer for next POI recommendation. In: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 1144–1153 (2022). https://doi.org/10.1145/3477495.3531983
23. Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C.: Gradient surgery for multi-task learning. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 5824–5836 (2020)
24. Zeng, J., He, X., Tang, H., Wen, J.: A next location predicting approach based on a recurrent neural network and self-attention. In: Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom). pp. 309–322. Springer (2019). https://doi.org/10.1007/978-3-030-30146-0_22